

CAPACITORS & INDUCTORS

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1. INTRODUCTION

Capacitors and inductors are passive elements, each of which has the ability to both store and deliver finite amount of energy. Unlike resistors, which dissipate energy, capacitors and inductors do not dissipate but store energy, which can be retrieved at a later time. For this reason, capacitors and inductors are called storage elements. They differ from ideal source in the respect, since they cannot sustain a finite average power flow over an infinite time interval. Although they are classed as linear elements, the current-voltage relationship for these elements is time-dependent, leading to many interesting circuits.

2.1 CAPACITORS

A capacitor is a passive element designed to store energy in its electric field. Besides resistors, capacitors are the most common electrical components. Capacitors are used extensively in electronics, communications, computers and power systems. For example, they are used in tuning circuits of radio receivers and as dynamic memory elements in computer systems.

A capacitor consists of two conducting plates separated by an insulator (or dielectric). The amount of potential difference present across the capacitor depends upon how much charge was deposited onto the plates by the work being done by the source voltage and also by how much capacitance the capacitor has and this is illustrated in Figure 1.

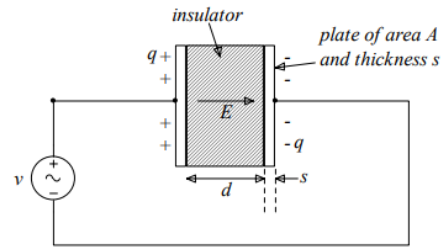


Figure 1. A capacitor model.

In many practical applications, the plates may be aluminum foil while the dielectric may be air, ceramic, paper, or mica. The amount of charge stored, represented by q , is directly proportional to the applied voltage v , so that

$$q = Cv \quad (1)$$

Where C is the constant of proportionality, is known as the capacitance of the capacitor. The unit of capacitance is the farad (F). From equation (1), we may derive the following definition- Capacitance is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F). [1 farad = 1 Coulomb/Volt]

Although the capacitance C of the capacitor is the ratio of the charge Q per plate to the applied voltage v , it does not depend on q or v . It depends on the physical dimensions of the capacitor. The capacitance is given by

$$C = \frac{A\epsilon}{d} \quad (2)$$

where A is the surface area of each plate, d is the distance between the plates, and ϵ is the permittivity of the dielectric material between the plates. Typically, capacitors have values on the picofarad (pF) to microfarad (μ F) range. Figure 2 shows the circuit symbols for fixed and variable capacitors.

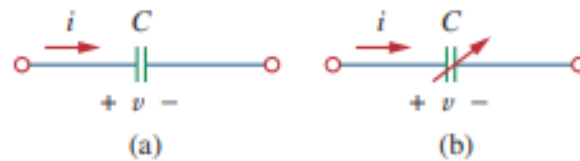


Figure 2. Circuit symbols for capacitors. (a) fixed capacitors, (b) variable capacitors.

To obtain the current-voltage relationship of the capacitor, we take the derivative of both sides of equation (1). Since

$$i = \frac{dq}{dt} \quad (3)$$

$$i = \frac{C dV}{dt} \quad (4)$$

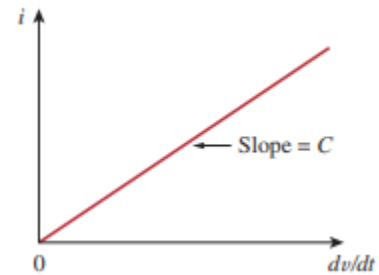


Figure 3. Current-voltage relationship of a capacitor.

The voltage-current relation of the capacitor can be obtained by integrating both sides of equation (4). We get

$$V = \frac{1}{C} \int_{-\infty}^t i dt \quad (5)$$

Or,

$$V = \frac{1}{C} \int_{t_0}^t i dt + v(t_0) \quad (6)$$

Where $v(t_0) = \frac{q(t_0)}{C}$ is the voltage across the capacitor at time t_0 .

Equation (6) shows that capacitor voltage depends on the past history of the capacitor current. Hence, the capacitor has memory – a property that is often exploited.

The instantaneous power delivered to the capacitor is

$$p = vi = Cv \frac{dv}{dt} \quad (7)$$

The energy stored in the capacitor is therefore

$$w = \frac{1}{2} Cv^2 = \frac{q^2}{2C} \quad (8)$$

Equation (8) represents the energy stored in the electric field that exists between the plates of the capacitor. This energy can be retrieved, since an ideal capacitor cannot dissipate energy.

We should note the following important properties of a capacitor:

1. Note from equation (4) that when the voltage across a capacitor is not changing with time (i.e. dc voltage), the current through the capacitor is zero. Thus,
A capacitor is an open circuit to dc.
2. The voltage on the capacitor must be continuous.
The voltage on a capacitor cannot change abruptly.
The capacitor resists an abrupt change in the voltage across it.
3. The ideal capacitor does not dissipate energy. It takes power from the circuit when storing energy in its field and returns previously stored energy when delivering power to the circuit.
4. A real, nonideal capacitor has a parallel-model leakage resistance. The leakage resistance may be as high as 100 M Ω and can be neglected for most practical applications.

Problem 1:

- (a) Calculate the charge stored on a 3-pF with 20 V across it.
- (b) Find the energy stored in the capacitor.

Solution

- (a) Since $q = Cv = 3 \times 10^{-12} \times 20 = 60 \text{ pC}$
- (b) The energy stored is $w = \frac{1}{2} Cv^2 = \frac{1}{2} \times 3 \times 10^{-12} \times 20^2 = 600 \text{ pJ}$

Problem 2:

What is the voltage across a 4.5- μF capacitor if the charge on one plate is 0.12 mC? How much energy is stored?

Answer 26.67 V, 1.6 mJ.

Problem 3:

The voltage across a 5- μF capacitor is $v(t) = 10 \cos 6000t \text{ V}$. Calculate the current through it.

Solution:

By definition, the current is

$$\begin{aligned} i &= C \frac{dv}{dt} = 5 \times 10^{-6} \times \frac{d}{dt} (10 \cos 6000t) \\ &= -5 \times 10^{-6} \times 10 \sin 6000t \\ &= -0.3 \sin 6000t \text{ A} \end{aligned}$$

Problem 4:

If a 10- μF capacitor is connected to a voltage source with

$$v(t) = 75 \sin 2000t \text{ V}$$

Determine the current through the capacitor.

Answer: $1.5 \cos 2000t \text{ A}$.

Problem 5:

Determine the voltage across a 2- μF capacitor if the current through it is $i(t) = 6 e^{-3000t} \text{ mA}$.

Assume that the initial capacitor voltage is zero.

Solution

Since $v = \frac{1}{C} \int_0^t i dt + v(0)$ and $v(0) = 0$.

$$\text{Hence } v = \frac{1}{2 \times 10^{-6}} \int_0^t 6e^{-3000t} \times 10^{-3} V = (1 - e^{-3000t})V$$

Problem 6:

The current through a 100- μF capacitor is $i(t) = 50 \sin 120\pi t \text{ mA}$. Calculate the voltage across it at $t = 1 \text{ ms}$ and $t = 5 \text{ ms}$. Take $v(0) = 0$.

Answer: 93.14 mV , 1.736 V .

Problem 7:

What is the voltage across a 3- μF capacitor if the charge on one plate is 0.12 mC ? How much energy is stored?

Answer: 40 V , 2.4 mJ .

Problem 8:

Obtain the energy stored in each capacitor in Figure 3(a) under dc conditions.

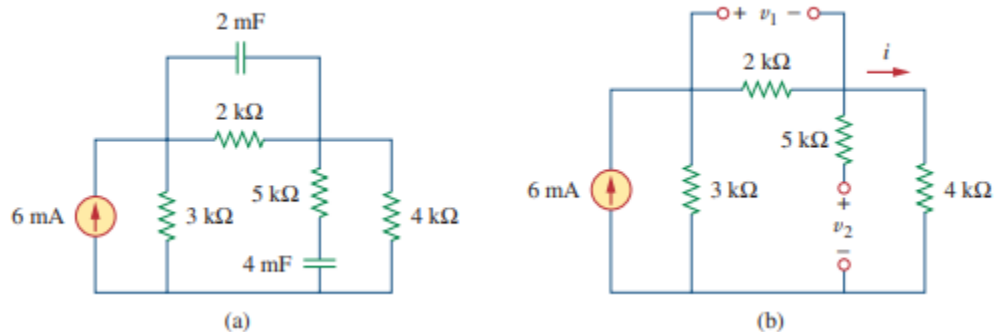


Figure 4.

Solution:

Under dc conditions, we replace each capacitor with an open circuit, as shown in Figure 3(b). The current through the series combination of the 2-k Ω and 4-k Ω resistors is obtained by current division as

$$i = \frac{3}{3 + 2 + 4}(6 \text{ mA}) = 2 \text{ mA}$$

Hence, the voltages v_1 and v_2 across the capacitors are

$$v_1 = 2000i = 4 \text{ V}$$

$$v_2 = 4000i = 8 \text{ V}$$

And the energies stored in them are

$$w_1 = \frac{1}{2}C_1v_1^2 = \frac{1}{2}(2 \times 10^{-3})(4^2) = 16 \text{ mJ}$$

$$w_2 = \frac{1}{2}C_2v_2^2 = \frac{1}{2}(4 \times 10^{-3})(8^2) = 128 \text{ mJ}$$

Problem 9:

Under dc conditions, find the energy stored in the capacitors in Figure 4.

Answer: 20.25 mJ, 3.375 mJ.

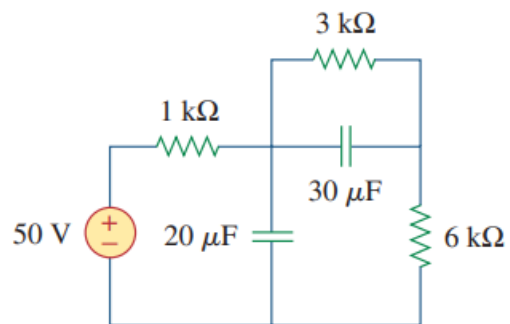


Figure 5.

Problem 10:

Calculate the energy stored in the capacitor of the circuit in Figure 5 under DC conditions.

Answer: 72 micro-joules.

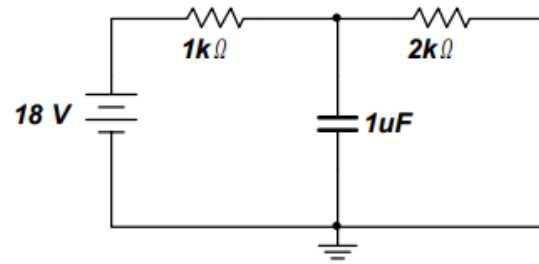


Figure 6.

Problem 11:

Calculate the energy stored in the capacitors of the following circuit shown in Figure 6(a) under DC conditions.

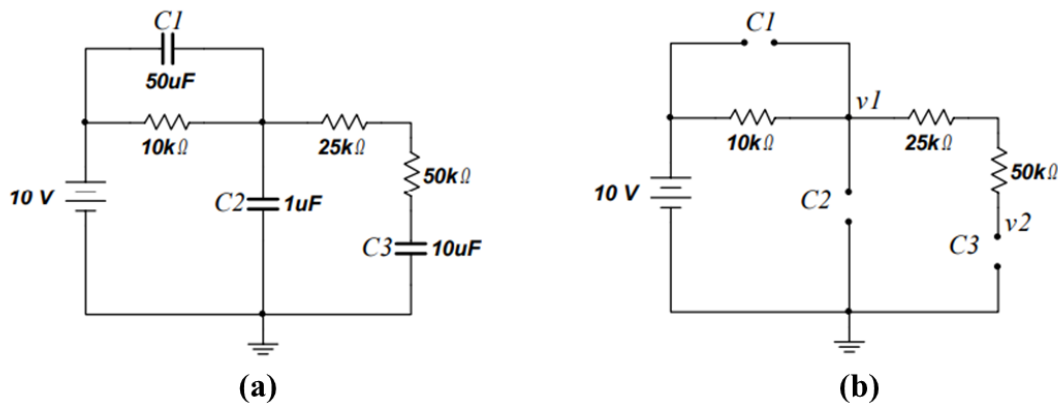


Figure 7.

Solution:

DC conditions imply that the capacitor behaves like an open circuit and the corresponding circuit is shown in Figure 6(b).

From this circuit we see that the voltages v_1 and v_2 are both equal to 10 V and thus the voltage across capacitor C_1 is 0 V.

For capacitor C_1 : $w_1 = 0 \text{ Joules}$

For capacitor C_2 : $w_2 = \frac{1}{2} C_2 v_1^2 = \frac{1}{2} (1 \times 10^{-6}) (10^2) = 50 \mu J$

For capacitor C_3 : $w_3 = \frac{1}{2} C_3 v_2^2 = \frac{1}{2} (10 \times 10^{-6}) (10^2) = 500 \mu J$

Problem 12:

Find the maximum energy stored in the capacitor of the following Figure 8 and the energy dissipated in the resistor over the interval $0 < t < 0.5$ s.

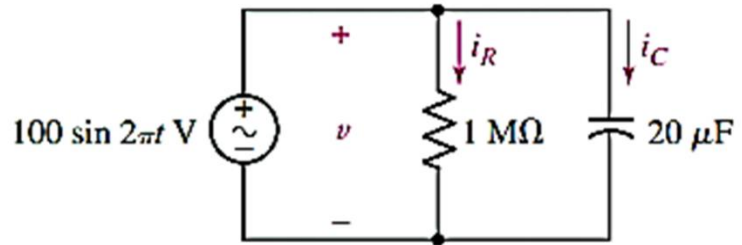


Figure 8.

Solution:

$$w_c(t) = \frac{1}{2} C v^2 = 0.1 \sin^2 2\pi t \text{ J}$$

$$P_R = \frac{V^2}{R} = \frac{10^2 \sin^2 2\pi t}{10^6} \text{ W}$$

$$w_R = \int_0^{0.5} P_R dt = \int_0^{0.5} \frac{10^2 \sin^2 2\pi t}{10^6} dt \text{ J}$$

Problem 13:

Find the voltage across and the charge on capacitor C_1 in Figure 9 after it has charged up to its final value.

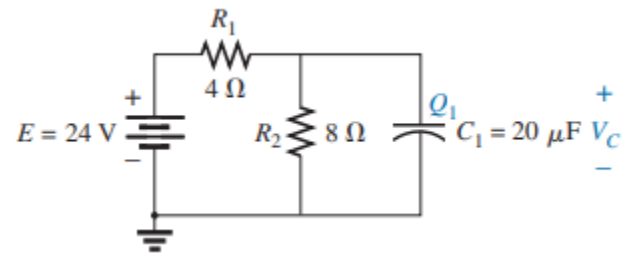


Figure 9.

Answer: 16V, 320 μC.

Problem 14:

Find the voltage across and the charge on each capacitor of the network in Figure 10 after each has charged up to its final value.

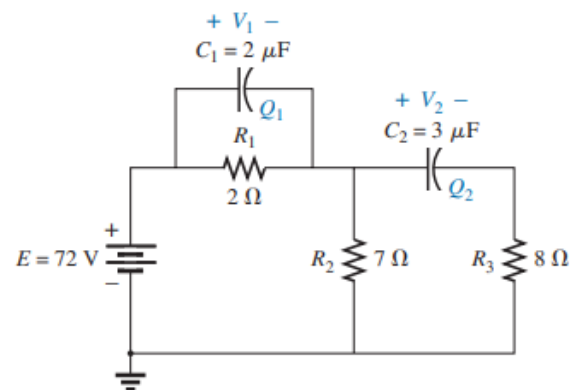


Figure 10.

Answer: 16 V, 56 V; 32 μC, 168 μC.

2.2 Series and Parallel Capacitors

We know from resistive circuits that series-parallel combination is a powerful tool for reducing circuits. This technique can be extended to series-parallel connections of capacitors, which are sometimes encountered. We desire to replace these capacitors by a single equivalent capacitor C_{eq} .

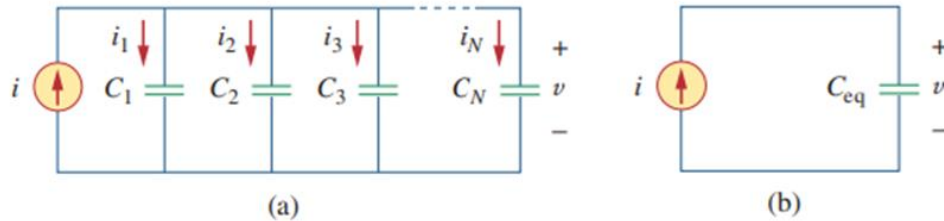


Figure 11. (a) Parallel-connected N capacitors, (b) equivalent circuit for the parallel capacitors.

First we obtain the equivalent capacitor C_{eq} of N capacitors in parallel,

$$C_{eq} = C_1 + C_2 + C_3 + \dots + C_N \quad (9)$$

The equivalent capacitance of N parallel-connected capacitors is the sum of the individual capacitances.

We observe that capacitors in parallel combine in the same manner as resistors in series.

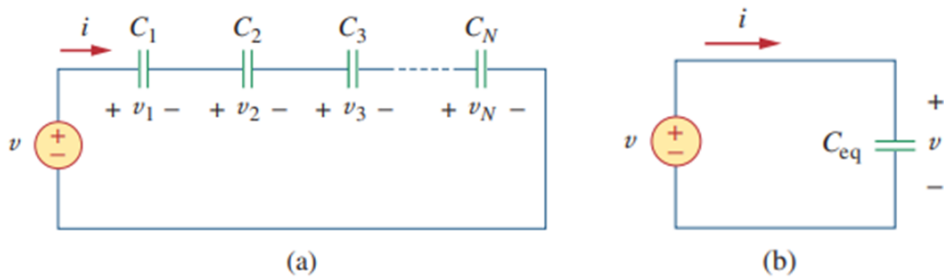


Figure 12. (a) Series-connected N capacitors, (b) equivalent circuit for the series capacitors.

Now we will obtain C_{eq} of N capacitors connected in series, where

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N} \quad (10)$$

The equivalent capacitance of series-connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.

Problem 15:

Find the equivalent capacitance seen between terminals a and b of the circuit in Figure 13.

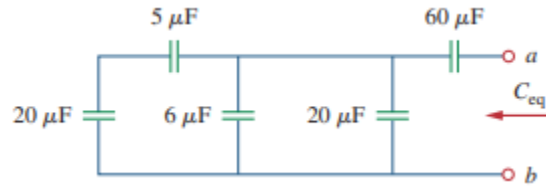


Figure 13.

Solution:

The 20-μF and 5- μF capacitors are in series; their equivalent capacitance is

$$\frac{20 \times 5}{20 + 5} = 4 \mu\text{F}$$

This 4-μF capacitor is in parallel with the 6-μF and 20-μF capacitors; their combined capacitance is

$$4 + 6 + 20 = 30 \mu\text{F}$$

This 30-μF capacitor is in series with the 60-μF capacitor. Hence, the equivalent capacitance for the entire circuit is

$$C_{eq} = \frac{30 \times 60}{30 + 60} = 20 \mu\text{F}$$

Problem 16:

Find C_{eq} for the network of Figure 14.

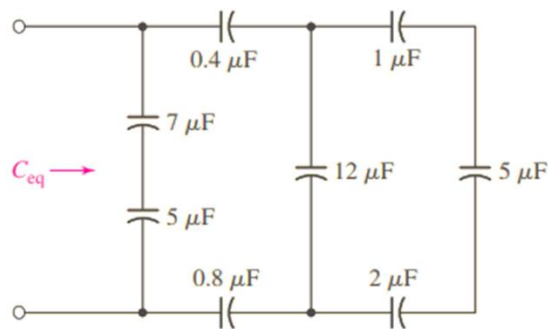


Figure 14.

Answer: 3.18 μF.

Problem 17:

Find the equivalent capacitance seen at the terminals of the circuit in Figure 15.

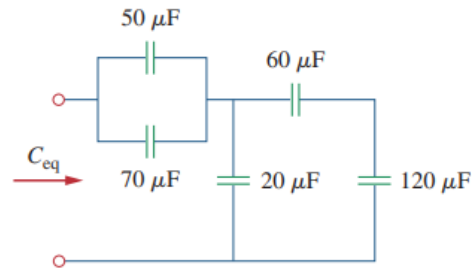


Figure 15.

Answer: 40- μF

Problem 18:

For the circuit in Figure 16:

- Find the total capacitance.
- Determine the charge on each plate.
- Find the voltage across each capacitor.

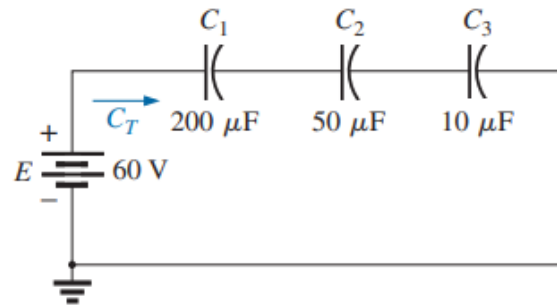


Figure 16.

a)

$$\begin{aligned}\frac{1}{C_T} &= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \\ &= \frac{1}{200 \times 10^{-6}} + \frac{1}{50 \times 10^{-6}} + \frac{1}{10 \times 10^{-6}} \\ &= 0.125 \times 10^6\end{aligned}$$

And $C_T = \frac{1}{0.125 \times 10^6} = 8 \mu F$

b) $Q_T = Q_1 = Q_2 = Q_3 = C_T V = (8 \times 10^{-6})(60) = 480 \mu F$

c)

$$V_1 = \frac{Q_1}{C_1} = \frac{480 \times 10^{-6} C}{200 \times 10^{-6} F} = 2.4 V$$

$$V_2 = \frac{Q_2}{C_2} = \frac{480 \times 10^{-6} C}{50 \times 10^{-6} F} = 9.6 V$$

$$V_3 = \frac{Q_3}{C_3} = \frac{480 \times 10^{-6} C}{10 \times 10^{-6} F} = 48 V$$

Problem 19:

For the network in Figure 17:

- Find the total capacitance.
- Determine the charge on each plate.
- Find the total charge.

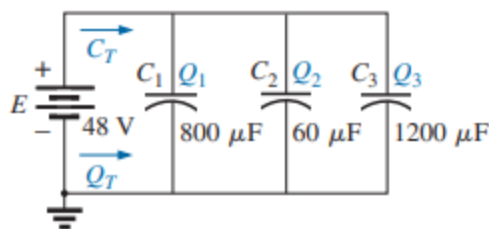


Figure 17.

Solution:

a)

$$C_T = C_1 + C_2 + C_3 = 800 \mu F + 60 \mu F + 1200 \mu F = 2060 \mu F$$

$$\text{b) } Q_1 = C_1 V = (800 \times 10^{-6} F)(48 V) = 38.4 mC$$

$$Q_2 = C_2 V = (60 \times 10^{-6} F)(48 V) = 2.88 mC$$

$$Q_3 = C_3 V = (1200 \times 10^{-6} F)(48 V) = 57.6 mC$$

$$\text{c) } Q_T = Q_1 + Q_2 + Q_3 = 38.4 mC + 2.88 mC + 57.6 mC = 98.88 mC$$

Problem 20:

Find the voltage across and the charge on each capacitor for the network in Figure 18(a).

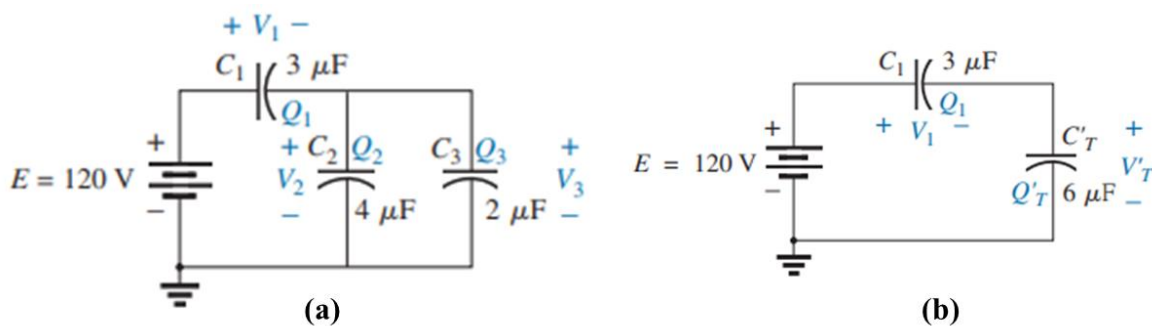


Figure 18.

Solution:

$$C'_T = C_1 + C_2 = 4 \mu F + 2 \mu F = 6 \mu F$$

$$C_T = \frac{C_1 C'_T}{C_1 + C'_T} = \frac{3 \times 6}{3 + 6} = 2 \mu F$$

$$Q_T = C_T V = (2 \times 10^{-6})(120) = 240 \mu C$$

An equivalent circuit (Figure 18(b)) has

$$Q_T = Q_1 = Q'_T$$

and, therefore $Q_1 = 240 \mu C$

$$\text{And } V_1 = \frac{Q_1}{C_1} = \frac{240 \times 10^{-6}}{3 \times 10^{-6}} = 80 V$$

$$Q'_T = 240 \mu C$$

$$\text{Therefore } V'_T = \frac{Q'_T}{C'_T} = \frac{240 \times 10^{-6}}{6 \times 10^{-6}} = 40 V$$

And

$$Q_2 = C_2 V'_T = (4 \times 10^{-6})(40) = 160 \mu C$$

$$Q_3 = C_3 V'_T = (2 \times 10^{-6})(40) = 80 \mu C$$

Problem 21:

For the circuit in Figure 19, find the voltage across each capacitor.

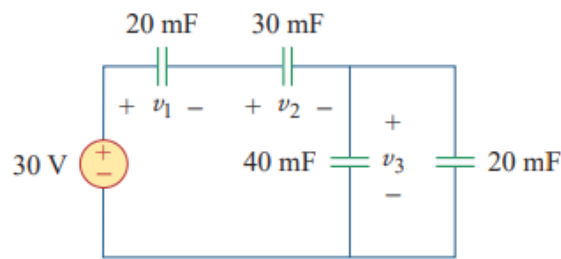


Figure 19.

Solution

40 mF and 20 mF capacitors are connected in parallel, their combined capacitance

$$C'_{eq} = 40 \text{ mF} + 20 \text{ mF} = 60 \text{ mF}$$

This 60 mF capacitor is in series with the 20 mF and 30 mF capacitors. Thus, the equivalent capacitance

$$C_{eq} = \frac{1}{\frac{1}{60} + \frac{1}{30} + \frac{1}{20}} \text{ mF} = 10 \text{ mF}$$

The total charge is

$$q = C_{eq} v = 10 \times 10^{-3} \times 30 = 0.3 C$$

This is the charge on the 20-mF, 30-mF and 60 mF capacitors, because they are in series with the 30-V source. Therefore,

$$v_1 = \frac{q}{C_1} = \frac{0.3}{20 \times 10^{-3}} = 15 \text{ V}$$

$$v_2 = \frac{q}{C_2} = \frac{0.3}{30 \times 10^{-3}} = 10 \text{ V}$$

$$v_3 = \frac{q}{C'_{eq}} = \frac{0.3}{60 \times 10^{-3}} = 5 \text{ V}$$

Problem 22:

Find the voltage across each of the capacitors in Figure 20.

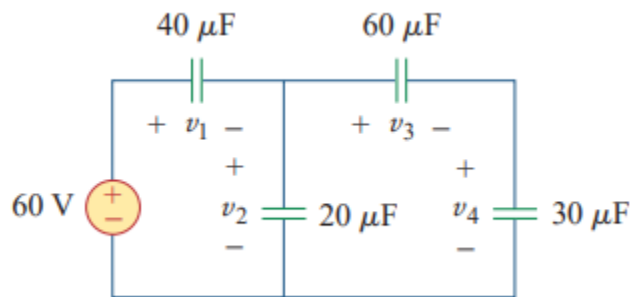


Figure 20.

Answer: 30V, 30V, 10V, 20V.

Problem 23:

For the circuit in Figure 21, determine:

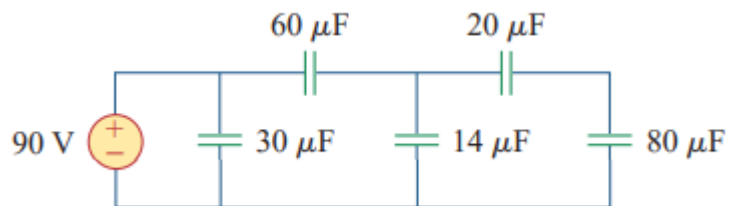


Figure 21.

- The voltage across each capacitor,
- The energy stored in each capacitor.

Solution: Try it yourself.

2.3 FIRST ORDER RC CIRCUIT

Now we have considered the three passive elements (resistors, capacitors, inductors), we are prepared to consider circuits that contain various combinations of two or three of the passive elements.

We carry out the analysis of RC and RL circuits by applying Kirchhoff's laws, as we did for resistive circuits. The only difference is that applying Kirchhoff's laws to purely resistive circuits results in algebraic equations, while applying the laws to RC and RL circuits produces differential equations, which are more difficult to solve than algebraic equations.

The differential equations resulting from analyzing RC and RL circuits are of the first order. Hence, the circuits are collectively known as first-order circuits.

A first-order circuit is characterized by a first-order differential equation.

2.4 The Source-Free RC Circuit

A source-free RC circuit occurs when its dc source is suddenly disconnected. The energy already stored in the capacitor is released to the resistors.

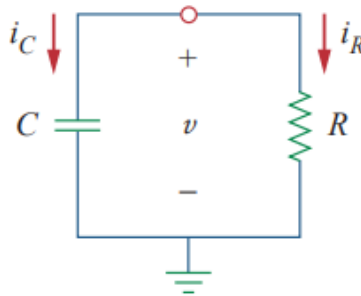


Figure 22.

Consider a series combination of a resistor and an initially charged capacitor, as shown in Figure 22. To determine the circuit response, we assume to be the voltage $v(t)$ across the capacitor. Since the capacitor is initially charged, we can assume that at time $t = 0$, the initial voltage is

$$v(0) = V_0 \quad (11)$$

With the corresponding value of the energy stored as

$$w(0) = \frac{1}{2} C V_0^2 \quad (12)$$

Applying KCL at the top node of the circuit in Figure 22.

$$i_C + i_R = 0 \quad (13)$$

By definition, $i_C = C \frac{dv}{dt}$ and $i_R = \frac{v}{R}$. Thus,

$$C \frac{dv}{dt} + \frac{v}{R} = 0 \quad (14)$$

This is a first-order differential equation, since only the first derivation of v is involved. After solve it, the capacitor voltage is

$$v(t) = V_0 e^{-\frac{t}{RC}} \quad (15)$$

This shows that the voltage response of the RC circuit is an exponential decay of the initial voltage. Since the response is due to the initial energy stored and the physical characteristics of the circuit and not due to some external voltage or current source, it is called the natural response of the circuit.

The natural response of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

The natural response is illustrated graphically in Figure 23. Note that $t = 0$, we have the current initial condition as in equation (11). As t increases, the voltage decreases toward zero. The rapidity with which the voltage decreases is expressed in terms of the *time constant*, τ .

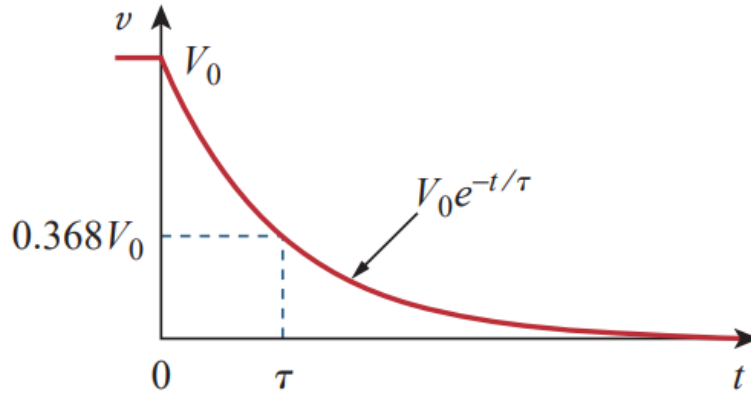


Figure 23, The voltage response of the RC circuit.

The time constant of a circuit is the time required for the response to decay by a factor of $1/e$ or 36.8 percent of its initial value.

This implies that at $t = \tau$, equation (15) becomes

$$V_0 e^{-\frac{t}{RC}} = V_0 e^{-1} = 0.368V_0$$

Or

$$\tau = RC \quad (16)$$

In terms of the time constant, Equation (15) can be written as

$$v(t) = V_0 e^{-\frac{t}{\tau}} \quad (17)$$

Observe from equation (16) that the smaller the time constant, the more rapidly the voltage decreases, that is, the faster the response. This is illustrated in Figure 24. A circuit with a small time constant gives a fast response in that it reaches the steady state (or final state) quickly due to quick dissipation of energy stored, whereas a circuit with a large time constant gives a slow response because it takes longer to reach steady state. At any rate, whether the time constant is small or large, the circuit reaches steady state in five-time constants.

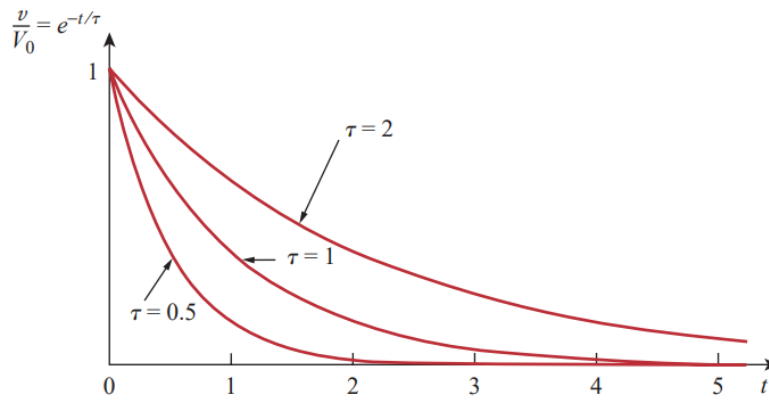


Figure 24. Plot of $\frac{v}{V_0} = e^{-\frac{t}{\tau}}$ for various values of the time constant.

With the voltage $v(t)$ in Equation (17), we can find the current $i_R(t)$,

$$i_R(t) = \frac{v(t)}{R} = \frac{V_0}{R} e^{-\frac{t}{\tau}} \quad (18)$$

The power dissipated in the resistor is

$$p(t) = v \cdot i_R(t) = \frac{V_0^2}{R} e^{-\frac{2t}{\tau}} \quad (19)$$

The energy absorbed by the resistor up to time t is

$$w_R(t) = \frac{1}{2} C V_0^2 \left(1 - e^{-\frac{2t}{\tau}} \right), \tau = RC \quad (20)$$

Notice the energy that was initially stored in the capacitor is eventually dissipated in the resistor. In finding the time constant $= RC$, R is often the Thevenin equivalent resistance at the terminals of the capacitor.

Problem 24: In Figure 25, let $v_C(0) = 15\text{ V}$, Find v_C , v_x , and i_x for $t > 0$.

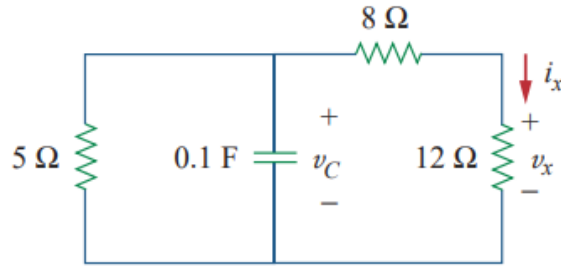


Figure 25.

Solution:

The 8Ω and 12Ω resistors in series can be combined to give $20\text{-}\Omega$ resistor. This $20\text{-}\Omega$ resistor in parallel with the $5\text{-}\Omega$ resistor can be combined so that the equivalent resistance is

$$R_{eq} = \frac{20 \times 5}{20 + 5} = 4\Omega$$

The time constant is

$$\tau = R_{eq}C = 4(0.1) = 0.4\text{ s}$$

Thus,

$$v = v(0)e^{-\frac{t}{\tau}} = 15e^{-\frac{t}{0.4}}\text{ V}$$

$$v_C = v = 15e^{-2.5t}\text{ V}$$

We can use voltage division to get v_x ; so

$$v_x = \frac{12}{12 + 8}v = 0.6(15e^{-2.5t}) = 9e^{-2.5t}\text{ V}$$

Finally,

$$i_x = \frac{v_x}{12} = 0.75e^{-2.5t}\text{ A}$$

Problem 25:

Refer to the circuit in Figure 26.

Let $v_C(0) = 45\text{ V}$. Determine v_C , v_x , and i_o for $t \geq 0$.

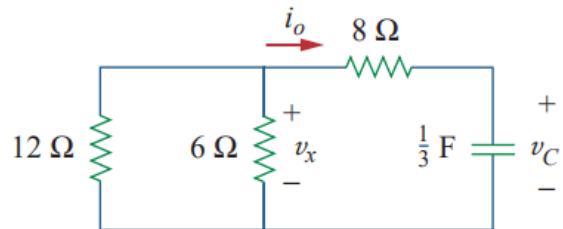


Figure 26.

Answer: $45e^{-0.25t}\text{ V}$, $15e^{-0.25t}\text{ V}$, $-3.75e^{-0.25t}\text{ A}$

Problem 26:

The switch in the circuit in Figure 27 has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.

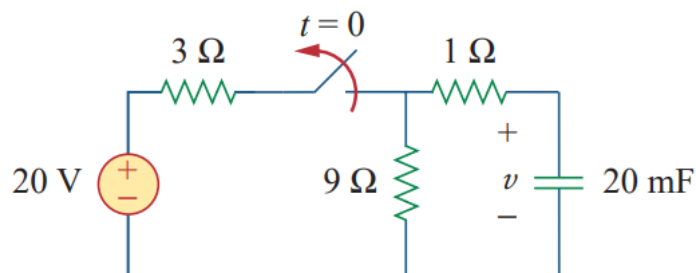


Figure 27.

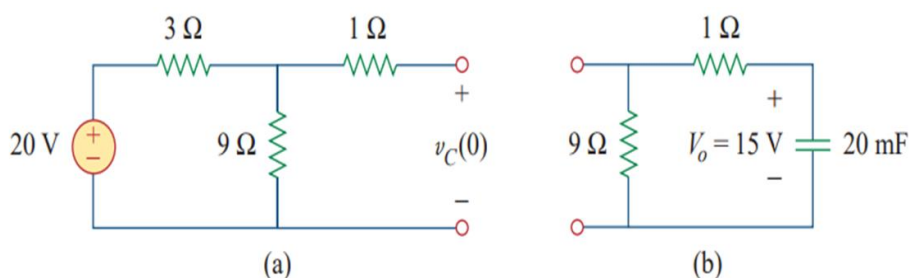
Solution:

Figure 28.

For $t < 0$, the switch is closed; the capacitor is an open circuit to dc, as represented in Figure 28(a). Using voltage division

$$v_C(t) = \frac{9}{9 + 3} (20) = 15V, t < 0$$

Since the voltage across a capacitor cannot change instantaneously, the voltage across the capacitor at $t = 0^-$ is the same at $t = 0$, or

$$v_C(0) = V_0 = 15V$$

For $t > 0$, the switch is opened, and we have the RC circuit shown in Figure 28(b). The 1-Ω and 9-Ω resistors in series give

$$R_{eq} = 1 + 9 = 10\Omega$$

The time constant is

$$\tau = R_{eq}C = 10 \times 20 \times 10^{-3} = 0.2 s$$

Thus, the voltage across the capacitor for $t \geq 0$ is

$$v(t) = v_c(0)e^{-\frac{t}{0.2}} = 15e^{-\frac{t}{0.2}} \text{ V}$$

$$\text{Or, } v(t) = 15e^{-5t} \text{ V}$$

The initial energy stored in the capacitor is

$$w_c(0) = \frac{1}{2} C v_c^2(0) = \frac{1}{2} \times 20 \times 10^{-3} \times 15^2 = 2.25 \text{ Joule}$$

Problem 27:

If the switch in Figure 29 opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $w_c(0)$.

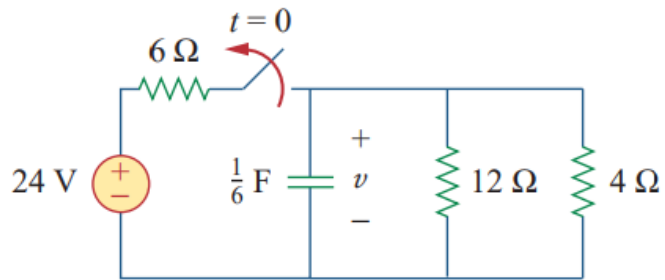


Figure 29.

Answer: $8e^{-2t} \text{ V}$, 5.33 Joule.

Problem 28: Using Figure 30, design a problem to help other students better understand source-free RC circuits.

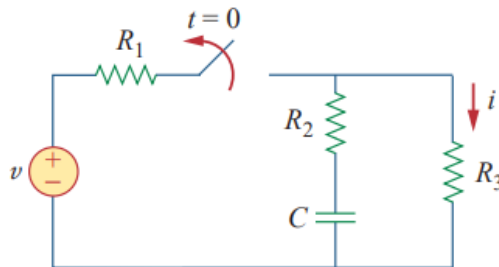


Figure 30.

Problem 29:

The switch in Figure 31 opens at $t = 0$. Find v_o for $t > 0$.

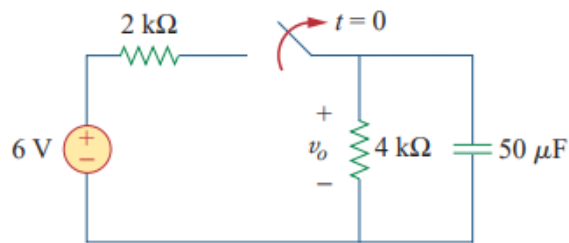


Figure 31.

Answer: $4e^{-\frac{t}{12}} \text{ V}$.

Problem 30:

For the circuit in Figure 32, find $v_o(t)$ for $t > 0$. Determine the time necessary for the capacitor voltage to decay to one-third of its value at $t = 0$.

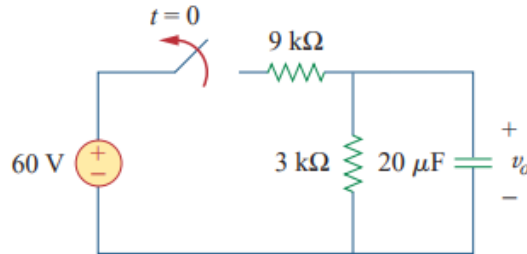


Figure 32.

Solution:

For $t < 0$,

$$V(0^-) = \frac{3}{3+9} (36V) = 9V$$

For $t < 0$, we have a source-free RC circuit

$$\tau = RC = 3 \times 10^3 \times 20 \times 10^{-6} = 0.06s$$

The voltage across the capacitor is

$$v(t) = 9e^{-16.667t}V$$

Let the time be t_0 .

$$3 = 9e^{-16.667t_0}$$

Or, $t_0 = 65.92ms$

3.1 INDUCTORS

An inductor is a passive element designed to store energy in its magnetic field. Inductors find numerous applications in electronic and power systems. They are used in power supplies, transformers, radios, TVs, radars, and electric motors.

Any conductor of electric current has inductive properties and maybe regarded as an inductor. But in order to enhance the inductive effect, a practical inductor is usually formed into a cylindrical coil with many turns of conducting wire, as shown in Figure 33.

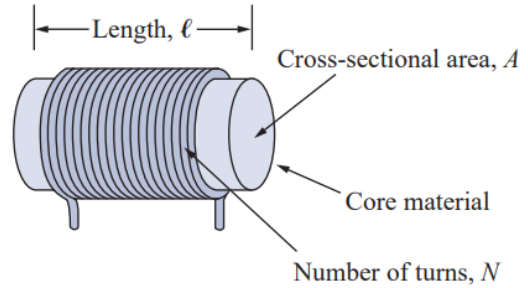


Figure 33. A typical form of inductor.

An inductor consists of a coil of conducting wire.

If current is allowed to pass through an inductor, it is found that the voltage across the inductor is directly proportional to the time rate of change of the current.

$$v = L \frac{di}{dt} \quad (2.1)$$

where L is the constant of proportionality called the inductance of the inductor. The unit of inductance is the henry (H).

Inductance is the property whereby an inductor exhibits opposition to the change of current flowing through it, measured in henrys (H).

The inductance of an inductor depends on its physical dimension and construction. For example, for the inductor, (solenoid) shown in Figure 33

$$L = \frac{N^2 \mu A}{l}$$

where N is the number of turns, l is the axial length, A is the cross-sectional area, and μ is the permeability of the medium.

The circuit symbols for inductors are shown in Figure 34.

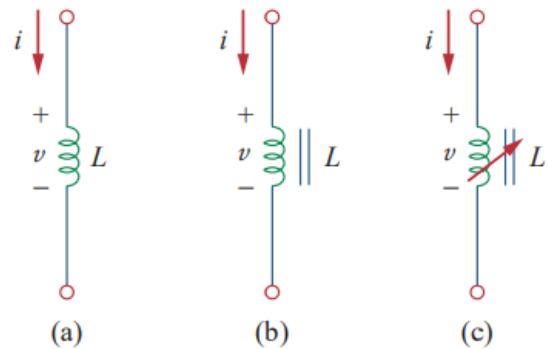


Figure 34. Circuit symbols for inductor: (a) air core, (b) iron core, (c) variable iron core.

The current-voltage relationship is obtained from Equation (2.1) as

$$di = \frac{1}{L} v dt$$

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0) \quad (2.2)$$

where $i(t_0)$ is the total current for $-\infty < t < t_0$ and $i(-\infty) = 0$.

The inductor is designed to store energy in its magnetic field. The energy stored can be obtained from Equation (2.1) and (2.2). The power delivered to the inductor is

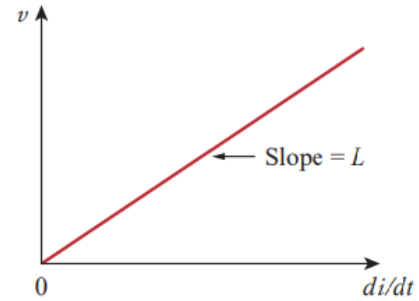


Figure 35. Voltage-current relationship of an inductor.

$$p = vi = \left(L \frac{di}{dt} \right) i \quad (2.3)$$

The energy stored is

$$w = \frac{1}{2} Li^2 \quad (2.4)$$

We should note the following important properties of an inductor.

1. The voltage across an inductor is zero when the current is constant. Thus,
An inductor acts like a short circuit to dc
2. An important property of the inductor is its opposition to the change in current flowing through it.
The current through an inductor cannot change instantaneously.
However, the voltage across an inductor can change abruptly.
3. Like the ideal capacitor, the ideal inductor does not dissipate energy. The energy stored in it can be retrieved at a later time.
4. A practical, nonideal inductor has a significant resistive component, as shown in Figure 35. This is due to the fact that the inductor is made of a conducting material such as copper, which has some resistance. This resistance is called the winding resistance, R_w , and it appears in series with the inductance of the inductor. The presence of R_w makes it both an energy storage device and an energy dissipation device.

3.2 Series and Parallel Inductors

Consider a series connection of N inductors, as shown in Figure 36(a), with the equivalent circuit shown in Figure 36(b). The inductors have the same current through them.

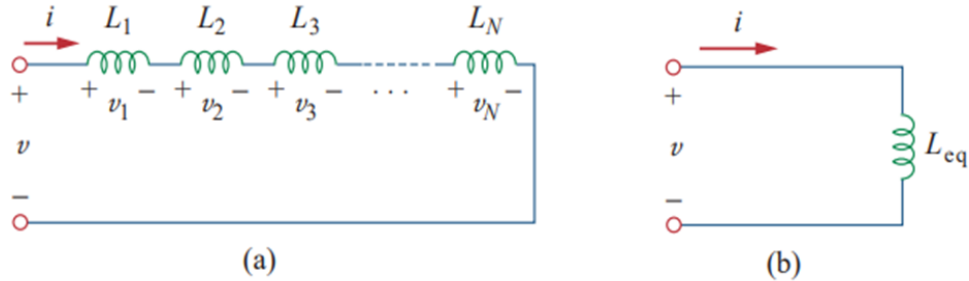


Figure 36. (a) A series connection of N inductors, (b) equivalent circuit for the series inductors.

The equivalent inductance of the series-connected set of inductors is

$$L_{eq} = L_1 + L_2 + L_3 + \dots + L_N \quad (2.5)$$

The equivalent inductance of series-connected inductors is the sum of the individual inductances. Inductors in series are combined in exactly the same way as resistors in series.

We now consider a parallel connection of N inductors, as shown in Figure 37(a), with the equivalent circuit of Figure 37(b). The inductors have the same voltage across them.

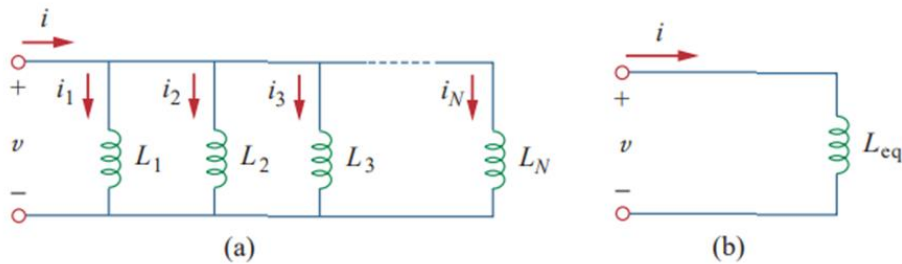


Figure 37. (a) A parallel connection of N inductors, (b) equivalent circuit for the parallel inductors.

The equivalent inductance of a parallel-connected set of inductors is

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N} \quad (2.6)$$

The equivalent inductance of parallel inductors is the reciprocal of the sum of the reciprocals of the individual inductances. Note that the inductors in parallel are combined in the same way as resistors in parallel.

Problem 31:

The current through a 0.1-H inductor is $i(t) = 10t e^{-5t}$ A. Find the voltage across the inductor and the energy stored in it.

Solution:

Since $v = L \frac{di}{dt}$ and $L = 0.1$ H,

$$v = 0.1 \frac{d}{dt}(10t e^{-5t}) = e^{-5t} + t(-5)e^{-5t} = e^{-5t}(1 - 5t) \text{ V}$$

The energy stored is

$$w = \frac{1}{2} Li^2 = \frac{1}{2} (0.1) 100t^2 e^{-10t} = 5t^2 e^{-10t} \text{ Joule}$$

Problem 32:

If the current through a 1-mH inductor is $i(t) = 20 \cos 100t$ mA, find the terminal voltage and the energy stored.

Answer: $-2 \sin 100t$ mV, $0.2 \cos^2 100t$ μ J.

Problem 33:

The current through a 10-mH inductor is $6e^{-\frac{t}{2}}$ A. Find the voltage and the power at $t = 3$ s.

Solution:

$$v = L \frac{di}{dt} = 10 \times 10^{-3} \times \frac{d}{dt} \left(6e^{-\frac{t}{2}} \right) = -30 e^{-\frac{t}{2}} \text{ mV}$$

$$v(3) = -30 e^{-\frac{3}{2}} \text{ mV} = -6.694 \text{ mV}$$

$$p = vi = -180 e^{-t} \text{ mW}$$

$$p(3) = -180 e^{-3} \text{ mW} = -8.962 \text{ mW}$$

Problem 34:

Find the current through a 5-H inductor if the voltage across it is

$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$$

Also, find the energy stored at $t = 5$ s. Assume $i(v) > 0$.

Solution:

Since $i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$ and $L = 5$ H

$$i = \frac{1}{5} \int_0^t 30t^2 dt + 0 = 2t^3 \text{ A}$$

The power $p = vi = 60 t^5$, and the energy stored is then

$$w = \int p dt = \int_0^5 60 t^5 dt = 156.25 \text{ kJ}$$

Alternatively, we can obtain the energy stored by

$$w = \frac{1}{2} Li(5)^2 - \frac{1}{2} Li(0)^2 = \frac{1}{2} L(2t^3)^2 - 0 = \frac{1}{2} L(2 \times 5^3)^2 = 156.25 \text{ kJ}$$

Problem 35:

The terminal voltage of a 2-H inductor is $v = 10(1 - t)$ V. Find the current flowing through it at $t = 4$ s and the energy stored in it at $t = 4$ s. Assume $i(0) = 2$ A.

Answer: -18 A, 320 Joule.

Problem 36:

The current through a 12-mH inductor is $4 \sin 100t$ A. Find the voltage, and also the energy stored in the inductor for $0 < t < \frac{\pi}{200}$ s.

Answer: $4.8 \cos 100t$ V, 96 mJ.

Problem 37:

The voltage across a 200-mH inductor is given by

$$v(t) = 3t^2 + 2t + 4 \text{ V for } t > 0.$$

Determine the current $i(t)$ through the inductor. Assume that $i(0) = 1$ A.

Answer: $5t^3 + 5t^2 + 20t + 1$ A

Problem 38:

Consider the circuit in Figure 38(a). Under dc conditions, find: (a) i , v_C , and i_L (b) the energy stored in the capacitor and inductor.

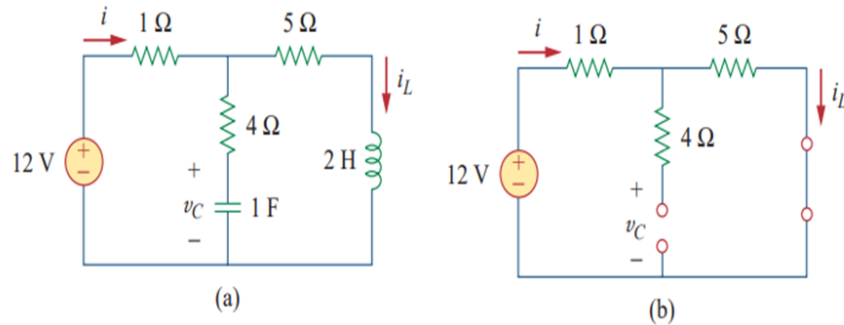


Figure 38.

Solution:

- a. Under dc conditions, we replace the capacitor with an open circuit and the inductor with a short circuit, as in Figure 38(b). It is evident from Figure 38(b) that

$$i = i_L = \frac{12}{1 + 5} = 2 \text{ A}$$

The voltage v_C is the same as the voltage across the 5-Ω resistor. Hence,

$$v_C = 5i = 10 \text{ V}$$

- b. The energy in the capacitor is

$$w_C = \frac{1}{2} C v_C^2 = \frac{1}{2} (1)(10^2) = 50 \text{ Joule}$$

And that in the inductor is

$$w_L = \frac{1}{2} L i_L^2 = \frac{1}{2} (2)(2^2) = 4 \text{ Joule}$$

Problem 39:

Determine v_C , i_L , and the energy stored in the capacitor and inductor in the circuit of Figure 39 under dc conditions.

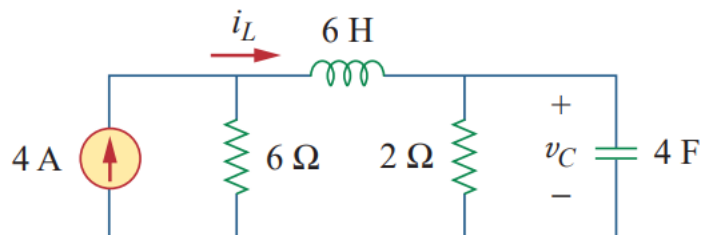


Figure 39.

Answer: 6 V, 3 A, 72 J, 27 J.

Problem 40: Find v_C , i_L , and the energy stored in the capacitor and inductor in the circuit of Figure 40 under dc conditions.

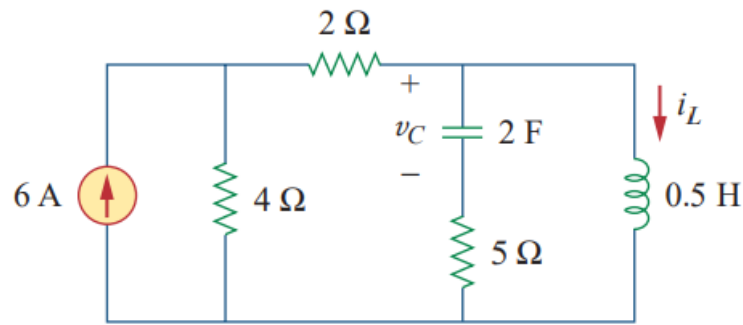


Figure 40.

Answer: 0 V, 2 A, 0 J, 1 J.

Problem 41: For the circuit in Figure 41, calculate the value of R that will make the energy stored in the capacitor the same as that in the inductor under dc conditions.

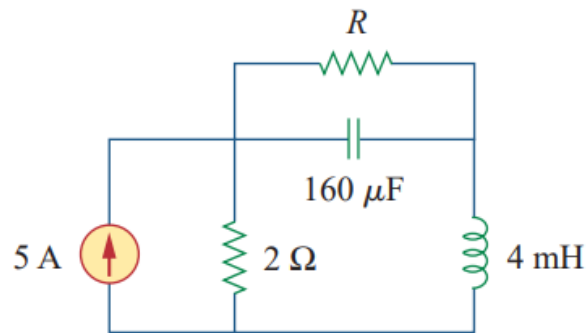


Figure 41.

Answer: 5 Ω.

Problem 42: Find the equivalent inductance of the circuit shown in Figure 42.

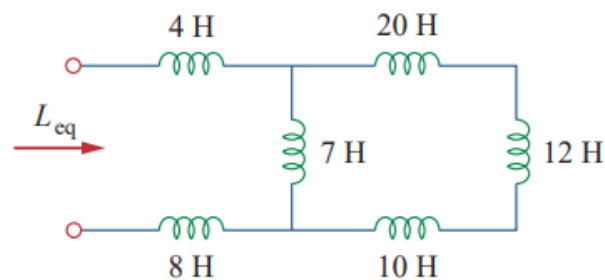


Figure 42.

Solution:

The 10-H, 12-H, and 20-H inductors are in series,; thus, combining them gives a 42-H inductance.

This 42-H inductor is in parallel with the 7-H inductor so that they are combined, to give

$$\frac{7 \times 42}{7 + 42} = 6 \text{ H}$$

This 6-H inductor is in series with the 4-H and 8-H inductors. Hence,

$$L_{eq} = 4 + 6 + 8 = 18 \text{ H}$$

Problem 43:

Calculate the equivalent inductance for the inductive ladder network in Figure 43.

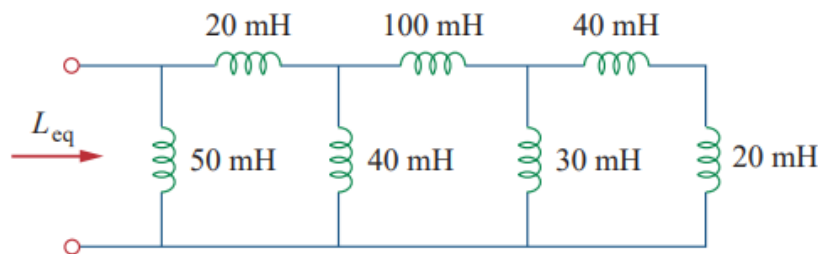


Figure 43.

Answer: 25 mH.

Problem 44:

Find the equivalent inductance looking into the terminals a-b of the circuit of Figure 44.

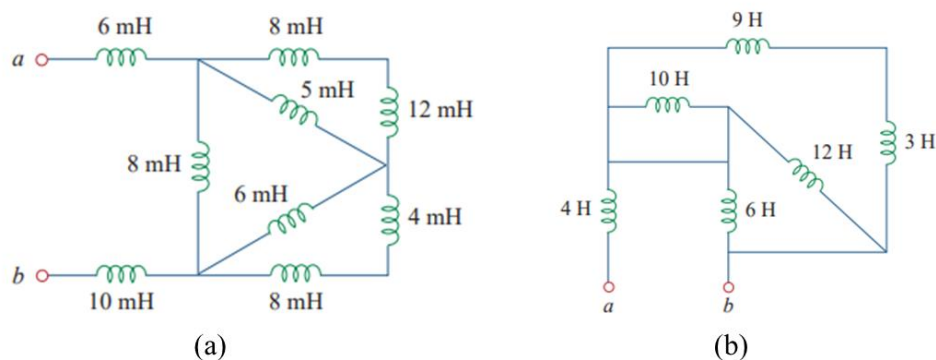


Figure 44.

Answer: (a) 20 mH, (b) 7 H.

3.3 The Source-Free RL Circuit

Consider the series connection of a resistor and an inductor, as shown in Figure 45. Our goal is to determine the circuit response, which we will assume to be the current $i(t)$ through the inductor. We select the inductor current as the response in order to take advantage of the idea that the inductor current cannot change instantaneously. At $t = 0$, we assume that the inductor has an initial current I_0 , or

$$i(0) = I_0$$

With the corresponding energy stored in the inductor as

$$w(0) = \frac{1}{2}LI_0^2 \quad (2.8)$$

Applying KVL around the loop in Figure ,

$$v_L + v_R = 0$$

But $v_L = L \frac{di}{dt}$ and $v_R = iR$.

Thus,

$$L \frac{di}{dt} + iR = 0 \quad (2.9)$$

Rearranging terms and integrating gives

$$i(t) = I_0 e^{-\frac{Rt}{L}} \quad (2.10)$$

This shows that the natural response of the RL circuit is an exponential decay of the initial current.

The current response is shown in Figure 46. The time constant for the RL circuit is

$$\tau = \frac{L}{R} \quad (2.11)$$

The smaller the time constant τ of a circuit, the faster the rate of decay of the response. The larger the time constant, the slower the rate of decay of the response. In one time constant the response has dropped to 36.8 percent of its initial value; the value of τ may also be determined graphically from this fact. At any rate, the response decays to less than 1 percent of its initial value (i.e. reaches steady state) after 5τ .

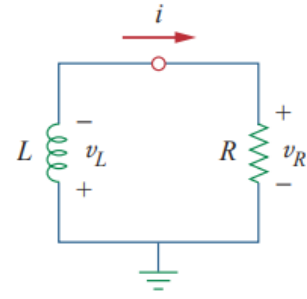


Figure 45. A source free RL circuit.

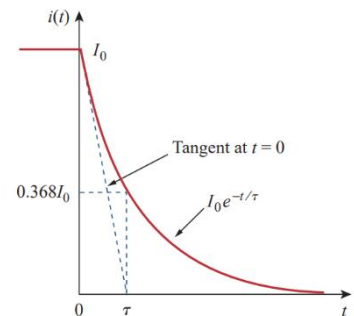


Figure 46. The current response of the RL circuit.

with τ again having the unit of seconds. Thus, Equation (2.10) may be written as

$$i(t) = I_0 e^{-\frac{t}{\tau}} \quad (2.12)$$

With the current in Eq. (2.12), we can find the voltage across the resistor as

$$v_R(t) = iR = I_0 R e^{-\frac{t}{\tau}} \quad (2.13)$$

The power dissipated in the resistor is

$$p = v_R \times i = I_0^2 R e^{-\frac{2t}{\tau}} \quad (2.14)$$

The energy absorbed by the resistor is

$$w_R(t) = \frac{1}{2} L I_0^2 (1 - e^{-\frac{2t}{\tau}}) \quad (2.14)$$

Problem 45:

Assuming that $i(0) = 10 \text{ A}$, calculate $i(t)$ and $i_x(t)$ in the circuit of Figure 47.

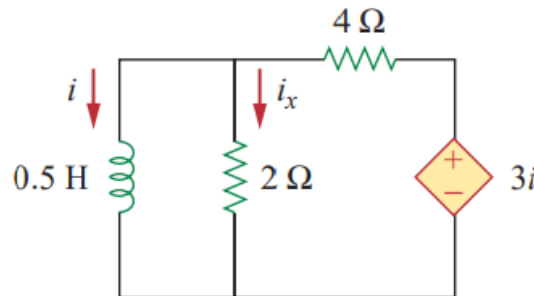


Figure 47.

Solution:

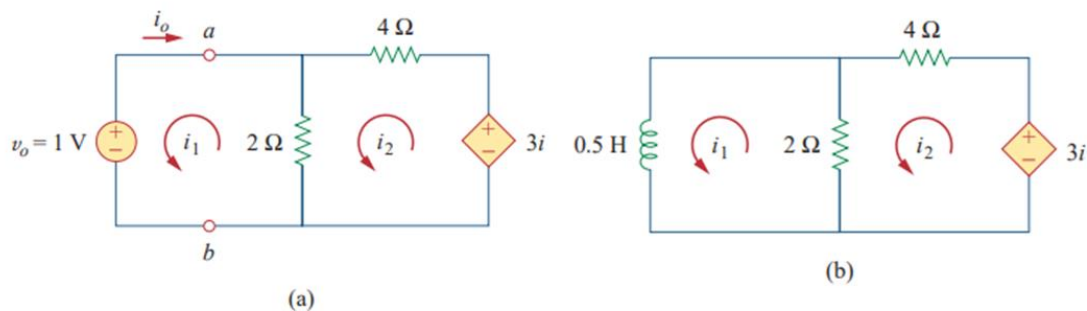


Figure 48.

The equivalent resistance is the same as the Thevenin resistance at the inductor terminals. Because of the dependent source, we insert a voltage source with $v_0 = 1V$ at the inductor terminals a-b, as in Figure 48(a). Applying KVL to the two loops results in

$$\begin{aligned} 2(i_1 - i_2) + 1 &= 0 \\ i_1 - i_2 &= -\frac{1}{2} \end{aligned} \quad \text{S.1}$$

And

$$\begin{aligned} 6i_2 - 2i_1 - 3i_1 &= 0 \\ i_2 &= \frac{5}{6} i_1 \end{aligned} \quad \text{S.2}$$

Substituting Eq. (S.2) into Eq. (S.1) gives

$$i_1 = -3 A, i_0 = -i_1 = 3 A$$

Hence,

$$R_{eq} = R_{Th} = \frac{v_0}{i_0} = \frac{1}{3} \Omega$$

The time constant is

$$\tau = \frac{L}{R_{eq}} = \frac{1/2}{1/3} = \frac{3}{2} s$$

Thus, the current through the inductor is

$$i(t) = i(0)e^{-\frac{t}{\tau}} = 10 e^{-(\frac{2}{3})t} A, t > 0$$

Alternate Method:

We may directly apply KVL to the circuit as in Figure 48(b).

For loop 1,

$$\begin{aligned} 0.5 \frac{di_1}{dt} + 2(i_1 - i_2) &= 0 \\ \frac{di_1}{dt} + 4i_1 - 4i_2 &= 0 \end{aligned} \quad \text{S.3}$$

For loop 2,

$$\begin{aligned} 6i_2 - 2i_1 - 3i_1 &= 0 \\ i_2 &= \frac{5}{6} i_1 \end{aligned} \quad \text{S.4}$$

Substituting Eq. (S.4) into Eq. (S.3) gives

$$\frac{di_1}{dt} + \frac{2}{3}i_1 = 0$$

Rearranging terms

$$\frac{di_1}{i_1} = -\frac{2}{3}dt$$

Since $i_1 = i$, we may replace i_1 with i and by integrating we can write

$$\ln \frac{i(t)}{i(0)} = -\frac{2}{3}t$$

Taking the powers of e , we finally obtain

$$i(t) = i(0)e^{-\left(\frac{2}{3}\right)t} = 10 e^{-\left(\frac{2}{3}\right)t} A, t > 0$$

The voltage across the inductor is

$$v = L \frac{di}{dt} = 0.5(10) \left(-\frac{2}{3}\right) e^{-\left(\frac{2}{3}\right)t} = -\frac{10}{3} e^{-\left(\frac{2}{3}\right)t} V$$

Since the inductor and the $2\text{-}\Omega$ resistor are in the parallel,

$$i_x(t) = \frac{v}{2} = -1.667 e^{-\left(\frac{2}{3}\right)t} A$$

Problem 46:

Find i and v_x in the circuit of Figure 49. Let $i(0) = 5 A$.

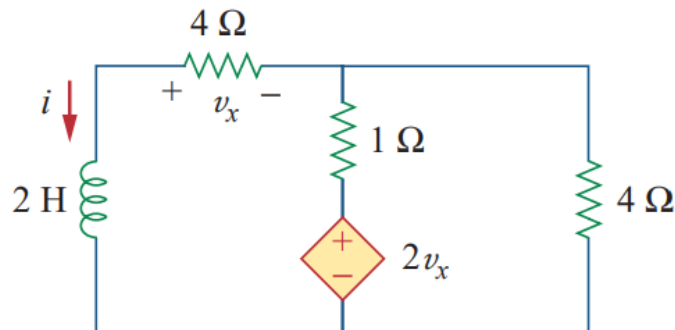


Figure 49.

Answer: $5 e^{-4t} V, -20 e^{-4t} V$.

Problem 47:

The switch in the circuit of Figure 49 has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.

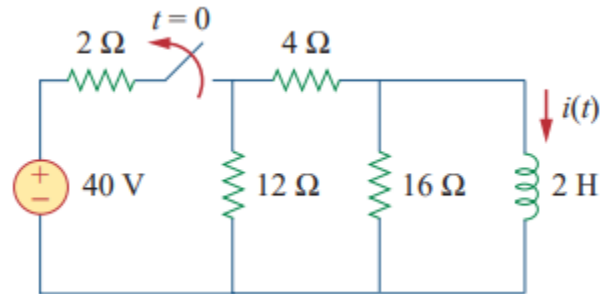


Figure 50.

Solution:

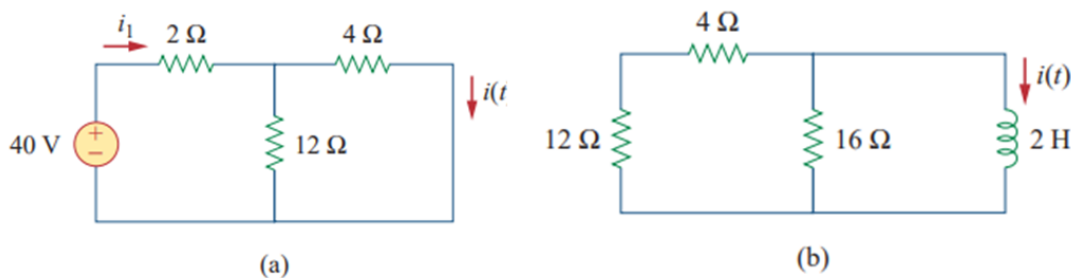


Figure 51. Solving the circuit of Figure 49: (a) for $t < 0$, (b) for $t > 0$.

When $t < 0$, the switch is closed, and the inductor acts as a short circuit to dc. The $16\text{-}\Omega$ resistor is short-circuited; the resulting circuit is shown in Figure 51(a). To get i_1 in Figure 51(a), we combine the $4\text{-}\Omega$ and $12\text{-}\Omega$ resistors in parallel to get

$$\frac{4 \times 12}{4 + 12} = 3\Omega$$

Hence,

$$i_1 = \frac{40}{2 + 3} = 8\text{ A}$$

We obtain $i(t)$ from i_1 in Figure 51 (a) using current division

$$i(t) = \frac{12}{12 + 4} i_1 = 6\text{ A}, t < 0$$

Since the current through an inductor cannot change instantaneously,

$$i(0) = i(0^-) = 6\text{ A}$$

When $t > 0$, the switch is open and the voltage source is disconnected. We now have the source-free RL circuit in Figure 51 (b). Combining the resistors, we have

$$R_{eq} = (12 + 4) \parallel 16 = 8 \Omega$$

The time constant is

$$\tau = \frac{L}{R_{eq}} = \frac{2}{8} = \frac{1}{4} \text{ s}$$

Thus,

$$i(t) = i(0)e^{-t/\tau} = 6 e^{-4t} \text{ A}$$

Problem 48: Find the time constant for each of the circuits in Figure 52.

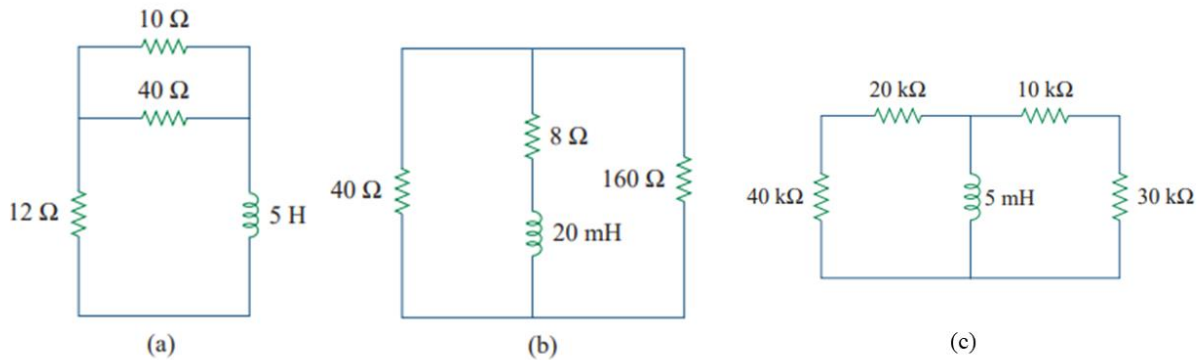


Figure 52.

Answer: (a) 0.25 s, (b) 0.5 ms, (c) 0.2083 μs .

Problem 49:

Consider the circuit of Figure 53. Find $v_o(t)$ if $i(0) = 2 \text{ A}$ and $v(t) = 0$.

Answer: $-2 e^{-16t} \text{ V}$.

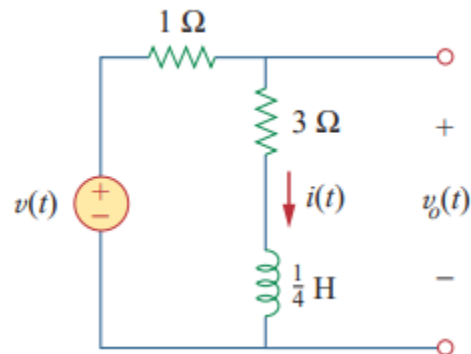


Figure 53.

Problem 50:

In the circuit of Figure 54, find $i(t)$ for $t > 0$ if $i(0) = 2 \text{ A}$.

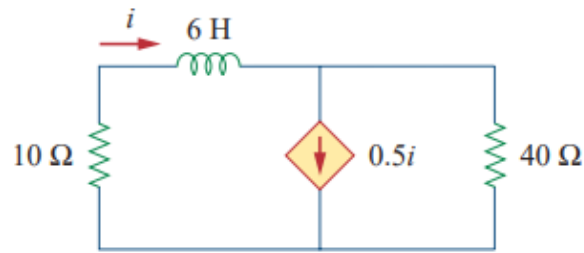


Figure 54.

Solution:

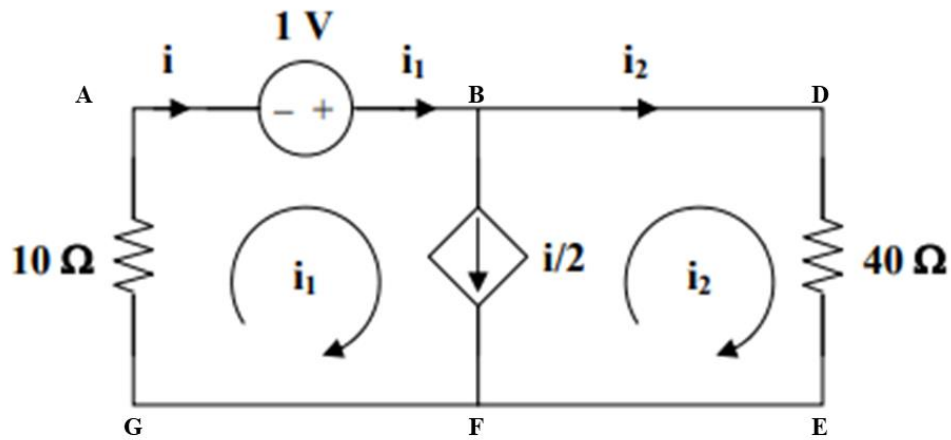


Figure 55.

To find R_{Th} we replace the inductor by a 1-V voltage source as shown in Figure 55.

Apply KVL at loop ABCDEFGA

$$10i_1 - 1 + 40i_2 = 0$$

Apply KCL at node B

$$i = i_2 + \frac{i}{2} \text{ and } i = i_1$$

$$\text{i.e. } i_1 = 2i_2 = i$$

Hence,

$$10i - 1 + 20i = 0$$

$$\text{Or, } i = \frac{1}{30}$$

Therefore,

$$R_{Th} = \frac{1}{i} = 30\Omega$$

$$\tau = \frac{L}{R_{Th}} = \frac{6}{30} = 0.2 \text{ s}$$

Hence,

$$i(t) = i(0)e^{-t/\tau} = 2 e^{-5t} \text{ A}$$